Vectors

ANSWER KEY

Vector arithmetic in three dimensions

1 Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$ and $|\mathbf{a}|$.

$\mathbf{a} + 2\mathbf{b} = (2, -1, 3) + (2, 8, -4) = (4, 7, -1)$. $|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$.

2 Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.

$|\mathbf{v}| = \sqrt{6^2 + (-3)^2 + 2^2} = \sqrt{49} = 7$. Unit vector: $\frac{1}{7}(6, -3, 2) = \left(\frac{6}{7}, -\frac{3}{7}, \frac{2}{7}\right)$.

3 Find $2\mathbf{a} - \mathbf{b}$ for $\mathbf{a} = (1, 3, -2)$ and $\mathbf{b} = (4, -1, 2)$.

$2\mathbf{a} - \mathbf{b} = (2, 6, -4) - (4, -1, 2) = (-2, 7, -6)$.

The scalar (dot) product and the angle between vectors

4 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 1(4) + (-2)(2) = -3 + 4 - 4 = -3$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, they are not perpendicular.

5 Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 2 - 2 + 4 = 4$. $|\mathbf{a}| = 3$, $|\mathbf{b}| = 3$. $\cos\theta = \frac{4}{9} \approx 0.4444$, so $\theta \approx 64^\circ$.

6 Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.

For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} = 0$:

$3(1) + k(-2) + (-2)(4) = 0 \Rightarrow 3 - 2k - 8 = 0 \Rightarrow -2k = 5 \Rightarrow k = -\frac{5}{2}$.

7 Find the angle between $\mathbf{a} = (3, 0, 4)$ and $\mathbf{b} = (0, 5, 0)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 3(0) + 0(5) + 4(0) = 0$. Since the dot product is zero, the vectors are perpendicular: the angle is 90°.

The vector equation of a line

8 Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.

$\mathbf{r} = (1, 2, -1) + t(2, -1, 3)$, where $t \in \mathbb{R}$.

9 Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.

Direction: $\vec{AB} = (5 - 2, 3 - 0, -2 - 1) = (3, 3, -3)$, or simplified $(1, 1, -1)$. Vector equation:

$\mathbf{r} = (2, 0, 1) + t(1, 1, -1)$.

10 Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.

Parallel lines share the same direction vector, so $\mathbf{r} = (0, 1, -2) + t(2, -1, 4)$.

Classifying lines: parallel, perpendicular and collinear points

11 Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.

Dot product: $1(2) + 2(1) + 2(-2) = 2 + 2 - 4 = 0$. Since the dot product is zero, the lines are perpendicular: the angle is 90°.

12 Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.

Direction vectors $(1, 1, 1)$ and $(2, 2, 2) = 2(1, 1, 1)$ are scalar multiples, so the lines are parallel. Testing whether $(2, 1, 0)$ lies on line 1: $(1 + s,\ s,\ 2 + s) = (2, 1, 0) \Rightarrow s = 1$ from the y-coordinate, but then the x -coordinate gives $1 + 1 = 2$ ✓ while the z-coordinate gives $2 + 1 = 3 \neq 0$ ✗. So the point does not lie on line 1 — the lines are parallel but distinct, and never meet.

13 Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.

$\vec{AB} = (3 - 1,\ 6 - 2,\ 7 - 3) = (2, 4, 4)$. $\vec{AC} = (5 - 1,\ 10 - 2,\ 11 - 3) = (4, 8, 8) = 2\vec{AB}$. Since $\vec{AC}$ is a scalar multiple of $\vec{AB}$ and both start at A, the three points lie on the same line, so A, B, C are collinear.

Intersecting and skew lines

14 Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.

Equate components: $2 + s = t$ (x); $1 = 2 + t \Rightarrow t = -1$ (y); then $s = t - 2 = -3$. Check the z-coordinate: line 1 gives $3 - s = 3 - (-3) = 6$; line 2 gives $5 - 2t = 5 - 2(-1) = 7$. Since $6 \neq 7$, the equations are inconsistent, and since the direction vectors $(1, 0, -1)$ and $(1, 1, -2)$ are not scalar multiples (not parallel), the lines are skew.

15 Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.

Equate components: $2 + s = 2t$ (x); $1 + s = 5 - t$ (y); $1 + 2s = 3 + t$ (z). From the y-equation, $s = 4 - t$. Substituting into the x-equation: $2 + (4 - t) = 2t \Rightarrow 6 - t = 2t \Rightarrow t = 2$, so $s = 2$. Checking the z-equation: $1 + 2(2) = 5$ and $3 + 2 = 5$ ✓. Point of intersection (using $s = 2$ in line 1):
$(2 + 2,\ 1 + 2,\ 1 + 4) = (4, 3, 5)$.

The equation of a plane

16 Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$. Equation: $2x - y + 3z = -3$.

17 Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.

$\vec{AB} = (-2, 3, 0)$, $\vec{AC} = (-2, 0, 4)$. Normal $= \vec{AB} \times \vec{AC} = (3 \cdot 4 - 0 \cdot 0,\ 0 \cdot (-2) - (-2) \cdot 4,\ (-2) \cdot 0 - 3 \cdot (-2)) = (12, 8, 6)$, simplifying to $(6, 4, 3)$. Plane: $6x + 4y + 3z = d$; substituting A: $6(2) = 12 = d$. Equation: $6x + 4y + 3z = 12$ (matching the intercept form $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$).

18 Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.

The plane has the form $3x - 2z = d$. Substituting the point: $3(1) - 2(2) = 3 - 4 = -1$. Equation: $3x - 2z = -1$.

19 Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.

$\vec{AB} = (-2, 1, 1)$, $\vec{AC} = (-1, -1, 3)$. Normal $= \vec{AB} \times \vec{AC} = (1 \cdot 3 - 1 \cdot (-1),\ 1 \cdot (-1) - (-2) \cdot 3,\ (-2) \cdot (-1) - 1 \cdot (-1)) = (4, 5, 3)$. Plane: $4x + 5y + 3z = d$; substituting A: $4(2) + 5(1) = 13 = d$. Equation: $4x + 5y + 3z = 13$.

Angles between lines, planes, and pairs of planes

20 Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.

Plane normal: $\mathbf{n} = (2, -1, 2)$. $\mathbf{d} \cdot \mathbf{n} = 2(2) + 1(-1) + 2(2) = 4 - 1 + 4 = 7$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = 3$. $\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{7}{9} \approx 0.778$, so $\theta \approx 51^\circ$.

21 Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$. $\cos\theta = \frac{|-1|}{6} \approx 0.167$, so $\theta \approx 80^\circ$.

Distances and same-side reasoning

22 Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

$d = \frac{|2(3) - 1(1) + 2(-2) - 6|}{\sqrt{4 + 1 + 4}} = \frac{|6 - 1 - 4 - 6|}{3} = \frac{5}{3}$.

23 Find the foot of the perpendicular from $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$, and the shortest distance from P to the line.

A general point on the line is $(1 + t, t, t)$. The vector from this point to P is $(3 - t, -t, 1 - t)$, which must be perpendicular to $(1, 1, 1)$: $(3 - t) - t + (1 - t) = 0 \Rightarrow 4 - 3t = 0 \Rightarrow t = \frac{4}{3}$. Foot of perpendicular: $\left(\frac{7}{3}, \frac{4}{3}, \frac{4}{3}\right)$. Vector from P to this foot: $\left(-\frac{5}{3}, \frac{4}{3}, \frac{1}{3}\right)$, with magnitude $\sqrt{\frac{25 + 16 + 1}{9}} = \sqrt{\frac{42}{9}} = \frac{\sqrt{42}}{3}$.

24 Determine whether the point $(1, 2, 3)$ lies on the same side of the plane $x + 2y - z = 4$ as the origin.

Substituting the point: $1 + 4 - 3 = 2$. Substituting the origin: $0 + 0 - 0 = 0$. Both values (2 and 0) are less than the plane's constant term (4), so the point $(1, 2, 3)$ lies on the same side of the plane as the origin.

25 Find the distance from the origin to the plane $3x + 4y = 20$ (which does not depend on z).
$d = \frac{|3(0) + 4(0) + 0(0) - 20|}{\sqrt{3^2 + 4^2 + 0^2}} = \frac{20}{5} = 4.$

A well-designed word problem: a drone's flight path

26 This question has four parts. A drone's position at time t seconds ($t \geq 0$) is $\mathbf{r}(t) = (2, 0, 10) + t(3, 4, -1)$, in metres, with z measured vertically. (a) Find the drone's position after 4 seconds. (b) The ground is the plane $z = 0$. Find the time at which the drone lands. (c) Find the drone's horizontal distance from the origin (in the xy-plane) at the moment it lands. (d) Find the angle between the drone's flight path and the ground, to the nearest degree.

(a) $\mathbf{r}(4) = (2 + 12,\ 0 + 16,\ 10 - 4) = (14, 16, 6)$. (b) The z-component is $10 - t$, so $10 - t = 0 \Rightarrow t = 10$ seconds. (c) At $t = 10$: position $= (2 + 30,\ 0 + 40,\ 10 - 10) = (32, 40, 0)$. Horizontal distance from the origin: $\sqrt{32^2 + 40^2} = \sqrt{1024 + 1600} = \sqrt{2624} = 8\sqrt{41} \approx 51.2$ m. (d) The flight direction is $(3, 4, -1)$ and the ground's normal is $(0, 0, 1)$. $\mathbf{d} \cdot \mathbf{n} = -1$. $|\mathbf{d}| = \sqrt{9 + 16 + 1} = \sqrt{26}$. $\sin\theta = \frac{1}{\sqrt{26}} \approx 0.196$, so $\theta \approx 11^\circ$.